

Panel MS-AR(1): common intercept and linear trend

1985–2019, excluding the US, Germany, Japan, and China

1 Model

Seasonally adjusted real GDP per worker, 2026-09-17 vintage, 1985Q1 through 2019Q4. United States, Germany, and Japan are excluded. China is not in this vintage. No country random effects and no catch-up term.

$$y_{it} = a + g t + z_{it}, \tag{1}$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \tag{2}$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}). \tag{3}$$

Three regimes. $E[z] = 0$ (the median regime mean is not pinned at 0). σ and ρ switch, with $|\rho| < 0.99$. Standard errors are delta-method from a numerical Hessian.

2 Common intercept and linear trend, 1985–2019, ex US, Germany, Japan, China

2026-09-17 SA panel, 1985Q1–2019Q4, excluding the US, Germany, and Japan. $y_{it} = a + gt + z_{it}$, $E[z] = 0$. 71 countries, 6378 observations, 1985-Q1–2019-Q4. Fitted countries: 71. Observations: 6378. Log-likelihood: 16926.37. Common intercept $a = -4.9832$ (0.0748). Common slope $g = 0.0062$ (0.0006).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-1.0619 (0.0909)	-0.7923 —	1.0563 (0.0728)
σ	0.0562 (0.0018)	0.0150 (0.0003)	0.0098 (0.0002)
ρ	0.9900 (0.0000)	0.9900 (0.0000)	0.9900 (0.0000)
$\Pi_0.$	0.8257 (0.0242)	0.1141 (0.0203)	0.0602 (0.0120)
$\Pi_1.$	0.0444 (0.0070)	0.9516 (0.0068)	0.0039 (0.0024)
$\Pi_2.$	0.0245 (0.0038)	0.0004 (0.0017)	0.9751 (0.0035)
π	0.1620 (0.0163)	0.3858 (0.0355)	0.4522 (0.0379)

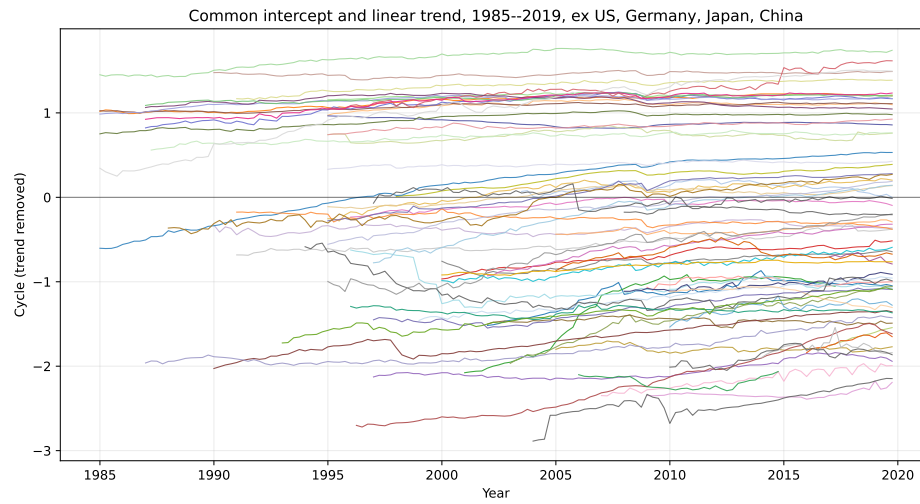


Figure 1: Country cycles after removing the common intercept and the linear trend.